

Name _____

Date _____

Total
28.5
/32

KNOWLEDGE

1. What is the exact radian measure of an angle of 225° ?

a) $\frac{20\pi}{12}$

b) $\frac{30\pi}{24}$

c) $\frac{48\pi}{14}$

d) $\frac{10\pi}{6}$

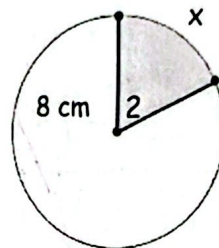
2. What is the missing quantity in the diagram?

a) 4 cm

b) 16 cm

c) 16 rad

d) 4 rad



$\theta = \frac{a}{r}$

$a = \theta r$

3. Determine the exact value of $\sec$ of $\frac{7\pi}{6}$.

a) $\frac{\sqrt{3}}{2}$

b) $-\frac{2\sqrt{3}}{3}$

c) $-\frac{1}{2}$

d) $-\frac{\sqrt{3}}{2}$

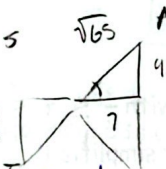
4. If $\cot \theta = \frac{7}{4}$, then what does $\cos \theta$ equal?

a) $\frac{4\sqrt{65}}{65}$

b) $\frac{4}{7}$

c) $\frac{7\sqrt{65}}{4}$

d) $-\frac{7\sqrt{65}}{65}$



has to be c thus value = $\frac{7\sqrt{65}}{65}$

5. Which of the following correctly express $4\sin\left(\frac{x}{6}\right)\cos\left(\frac{x}{6}\right)$ as a single sine or cosine function?

a) $4\sin\left(\frac{x}{6}\right)$

b) $2\sin\left(\frac{x}{3}\right)$

c) $4\sin\left(\frac{x}{3}\right)$

d) $2\sin\left(\frac{x}{6}\right)$

6. Determine the angle of $\csc \theta = -\sqrt{2}$ for the domain $0 \leq \theta \leq 4\pi$. Must include a diagram for full marks.

$\theta_1 = \frac{5\pi}{4}$

$\frac{1}{\sin} = -\frac{\sqrt{2}}{1}$

$\theta_2 = \frac{7\pi}{4}$

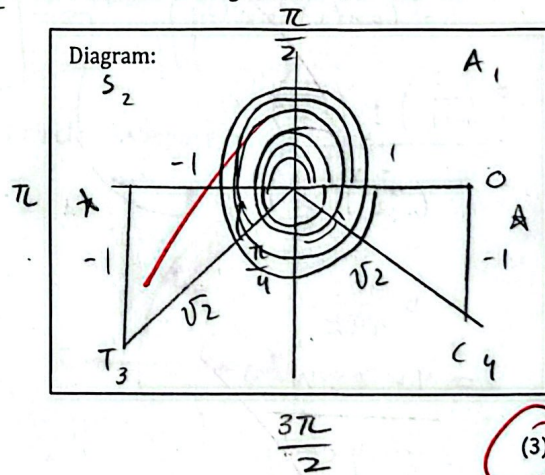
$\sin = \frac{1}{\sqrt{2}}$ RAA: $\frac{\pi}{4}$

$\theta_3 = \frac{15\pi}{4}$

$\theta_4 = \frac{13\pi}{4}$

$\sin \theta = \frac{1}{\sqrt{2}}$

$\theta = -\frac{\sqrt{2}}{2}$



7. The ceiling fan in your bedroom rotates 320 revolutions every 4 min. Determine the angular velocity of the ceiling fan in radians per second. Express your answer in exact and approximate form. Round to 2 decimal places when necessary.

$$\frac{320 \text{ rev}}{4 \text{ min}} \times \frac{2\pi}{1 \text{ rev}} \times \frac{1 \text{ min}}{60 \text{ sec}} = \frac{8\pi}{3}$$

Exact: $\frac{8\pi}{3} \text{ rad/sec}$

Approximate: 8.38 rad/sec

APPLICATION

1. Use the appropriate compound angle formula to determine the exact value of $\tan\left(-\frac{7\pi}{12}\right)$. Include a diagram for full marks. Final answer must be rationalized.

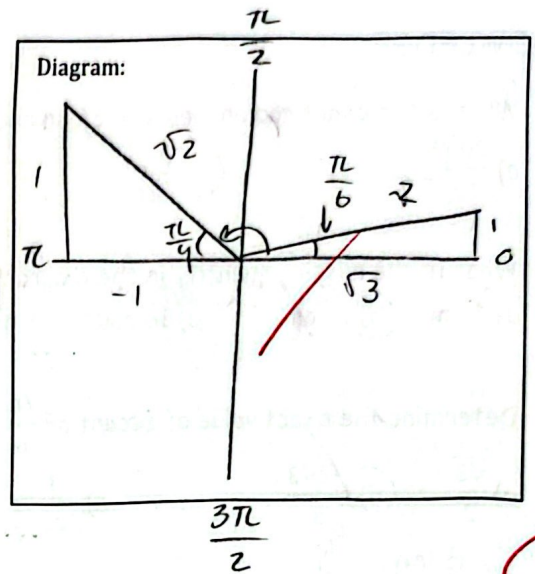
$$\tan\left(-\frac{7\pi}{12}\right) \rightarrow \frac{\pi}{6} - \frac{3\pi}{4}$$

$$= \tan(a-b)$$

$$\tan\left(\frac{\pi}{6} - \frac{3\pi}{4}\right) = \frac{\tan\left(\frac{\pi}{6}\right) - \tan\left(\frac{3\pi}{4}\right)}{1 + \tan\left(\frac{\pi}{6}\right)\tan\left(\frac{3\pi}{4}\right)}$$

$$\tan\left(\frac{\pi}{6} - \frac{3\pi}{4}\right) = \frac{\left(\frac{1}{\sqrt{3}}\right) - (-1)}{1 + \left(\frac{1}{\sqrt{3}}\right)(-1)}$$

$$= \frac{1 + \sqrt{3}}{2 + \sqrt{3}}$$



(4)

2. Given $\sec x = \frac{13}{-12}$ and $\cos y = \frac{-3}{5}$ with $\frac{\pi}{2} \leq x \leq \pi$ and $\pi \leq y \leq \frac{3\pi}{2}$, evaluate the following. Must include diagrams and answer must be fully simplified.

a. $\tan(x+y)$

$$\sec x = \frac{13}{-12}$$

$$\cos y = \frac{-3}{5}$$

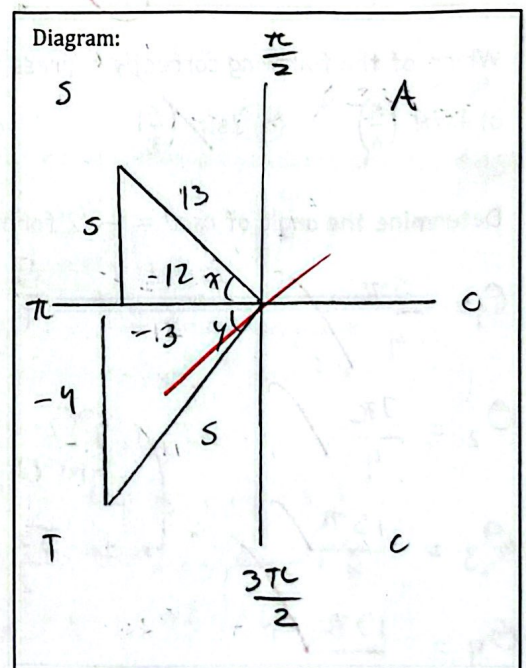
$$= \tan(x+y)$$

$$= \frac{\tan(x) + \tan(y)}{1 - \tan(x)\tan(y)}$$

$$= \frac{\left(\frac{5}{-12}\right) + \left(\frac{4}{3}\right)}{1 - \left(\frac{5}{-12}\right)\left(\frac{4}{3}\right)}$$

$$\cos = \frac{-12}{13}$$

$$= \frac{33}{56}$$



(4)

b. $\frac{\cos 2x}{\sin 2y}$

$$= \frac{1 - 2\sin^2 x}{2\sin y \cos y}$$

$$= \frac{1 - 2\left(\frac{5}{13}\right)^2}{2\left(-\frac{4}{5}\right)\left(-\frac{3}{5}\right)}$$

$$= \frac{2975}{4056}$$

(3) Prove the identity. Show all steps.

$$\frac{2\cos 2x}{\sin 2x - 2\sin^2 x} = 1 + \cot x$$

RS: $1 + \cot x$

$$= 1 + \frac{\cos x}{\sin x}$$

$$= \frac{\sin x + \cos x}{\sin x}$$

LS: $\frac{2\cos 2x}{\sin 2x - 2\sin^2 x}$

$$= \frac{2(\cos^2 x - \sin^2 x)}{2\sin x \cos x - 2\sin^2 x}$$

$$= \frac{2(\cos^2 x - \sin^2 x)}{2\sin x(\cos x - \sin x)}$$

$$= \frac{2(\cos x + \sin x)(\cos x - \sin x)}{2\sin x(\cos x - \sin x)}$$

$$= \frac{2(\cos x + \sin x)}{2\sin x}$$

$$= \frac{\cos x + \sin x}{\sin x}$$

$$= 1 + \cot x$$

COMMUNICATION

① Derek finds that the SIN key on his calculator is not working. Use a cofunction identity to determine another way to evaluate $\sin\left(\frac{8\pi}{9}\right)$.

0.342

Q2: $\sin\left(\frac{\pi}{2} + x\right) = \cos x$

$$\sin\left(\frac{8\pi}{9}\right) = -\cos x$$

$$\sin\left(\frac{\pi}{2} + x\right) = -\cos x$$

$$\frac{\pi}{2} + x = \frac{8\pi}{9}$$

$$x = \frac{7\pi}{18}$$

$$= -\cos\left(\frac{7\pi}{18}\right)$$

therefore, he can use the cofunction identity $\sin\left(\frac{\pi}{2} + x\right) = \cos x$ to find $\sin\left(\frac{8\pi}{9}\right)$ as they have the same value.

2. Explain why $\sin\left(\frac{9\pi}{7}\right) = -\sin\left(\frac{5\pi}{7}\right)$. Use a diagram of the unit circle to support your answer.

The RAA of $\sin\left(\frac{9\pi}{7}\right)$ and $-\sin\left(\frac{5\pi}{7}\right)$

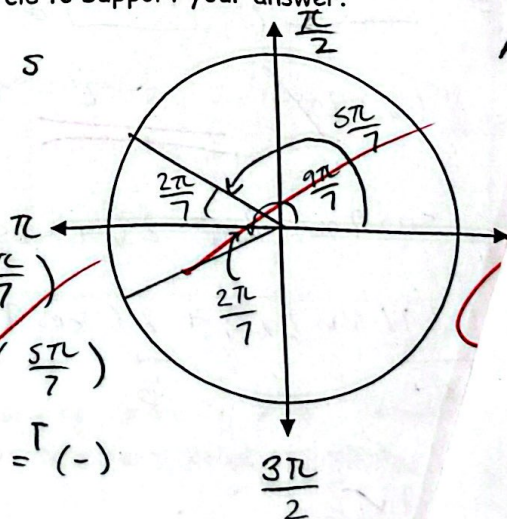
are the same $\left(\frac{2\pi}{7}\right)$ and $\sin\left(\frac{9\pi}{7}\right)$

is in Q3 so $\sin$ is $(-)$ and $-\sin\left(\frac{5\pi}{7}\right)$

is in Q2 so $\sin$ is $(+)$ but $-\sin\left(\frac{5\pi}{7}\right)$

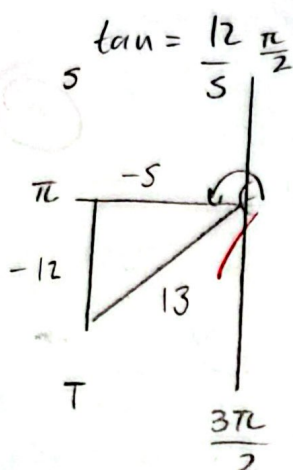
is being multiplied by $(-)$ and $(-)(+) = (-)$

so they have the same value.



THINKING

1. If $\tan x = \frac{12}{5}$, $\pi \leq x \leq \frac{3\pi}{2}$, determine an exact value for $\sin 4x$. Include a diagram for full marks.



$$\sin \theta = \frac{-12}{13}$$

$$\cos \theta = \frac{-5}{13}$$

$$\begin{aligned} &= \sin(2\theta + 2\theta) \\ &= \sin(2\theta) \cos(2\theta) + \cos(2\theta) \sin(2\theta) \\ &= 2\left(\frac{-12}{13}\right) 2\left(\frac{-5}{13}\right) + 2\left(\frac{-5}{13}\right) 2\left(\frac{-12}{13}\right) \\ &= \frac{240}{169} + \frac{240}{169} \\ &= \frac{480}{169} \end{aligned}$$

$$\sin 4x = \sin(2x + 2x)$$

$$\begin{aligned} &= \sin 2x \cos 2x + \cos 2x \sin 2x \\ &= \sin 2x (\cos 2x + \cos 2x) \\ &= 2 \sin x \cos x (2 \cos 2x) \\ &= 2 \sin x \cos x (2(2 \cos^2 x - 1)) \\ &= 2\left(\frac{-12}{13}\right)\left(\frac{-5}{13}\right)(2)\left(2\left(\frac{-5}{13}\right)^2 - 1\right) \end{aligned}$$

2. The expression below can be written in the form $a \sin 2x + b \cos 2x + c$. Determine the values of a , b and c .

$$-2 \sin^2 x \cos x - 4 \sin^2 x$$

$$= -2 \sin x \cos x - 4 \sin^2 x$$

$$= -(\sin 2x) - 2 + 2 - 4 \sin^2 x$$

$$= -\sin 2x + 2 + 2(1 - 2 \sin^2 x)$$

$$= (-1)(\sin 2x) + 2(\cos 2x - 1) + 2$$

$$a = -1$$

$$b = 2$$

$$c = 2$$

$$= 4\left(\frac{-12}{13}\right)\left(\frac{-5}{13}\right)\left(\frac{-119}{169}\right)$$

$$= \frac{-28560}{28561}$$

$$= -(2 \sin x \cos x) - 4 \sin^2 x$$

$$= -\sin 2x - 4 \sin^2 x$$

$$= -\sin 2x - 4(1 - \cos^2 x)$$

$$= -\sin 2x + 4(\cos^2 x - 1)$$

$$= -\sin 2x + 4 \cos^2 x - 4$$

$$= -\sin 2x + 2(2 \cos^2 x - 1 - 1)$$

$$= -\sin 2x + 2(\cos 2x - 1)$$

$$= -\sin 2x + 2 \cos 2x - 2$$

$$\begin{aligned} a &= -1 \\ b &= 2 \\ c &= -2 \end{aligned}$$